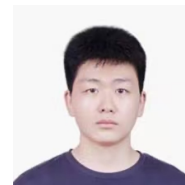


Ranjun Peng

B.Sc. in Artificial Intelligence | Research interests: LLMs & AI for Software Engineering

+86 18310886970 | 1230024861@student.must.edu.mo | [davidpengrj.github.io](https://github.com/davidpengrj)

Highlights: 4 publications, including a first-author paper under review at ACM TOSEM (CCF-A).



Education

Macau University of Science and Technology

Sep 2023 – Jun 2027 (expected)

B.Sc. in Artificial Intelligence | GPA: 3.3/4.0

Core courses: Machine Learning, Deep Learning, Data Mining, Computer Vision, Data Structures & Algorithms, Database Systems.

Publications & Research

Understanding Software Defect Prediction: A Large-scale Empirical Study Across Uncertainty Quantification and Performance Evaluation First author | ACM TOSEM (CCF-A), 2026 – under review

- Designed and ran a large-scale empirical study: 16 representative ML classifiers across 36 within-project and 32 cross-project datasets, with 5 uncertainty / 6 performance / 3 calibration metrics over 30 seeds; built the full pipeline (COD-clustering model selection, training/calibration, correlation analysis) and showed probability-based uncertainty is highly context-dependent and transfers poorly across projects.

Data Feature-Constrained GAN Data Generation for NIR Spectral Predictive Model Calibration

First author | 36th Chinese Process Control Conference (CPCC 2025)

- Proposed DFC-GAN, embedding the PCA principal-component covariance matrix into the generator loss as a statistical constraint; implemented the full architecture and custom loss, cutting downstream SVR/LSTM prediction RMSE by up to 50% on a public pharmaceutical NIR dataset.

Dictionary Learning-Based Knowledge Fusion ESN Hybrid Modeling

Second author | Industrial & Engineering Chemistry Research (ACS), published

- Co-designed and implemented the core ML algorithms (dictionary learning with sparse coding, knowledge-constrained ESN reservoir optimization); reproduced and trained all baselines (LSTM, SVR, Transformer-based Autoformer) on simulated and real chemical-plant data. DOI: [10.1021/acs.iecr.5c01050](https://doi.org/10.1021/acs.iecr.5c01050)

Towards Hybrid Modeling of Process Industries with Dynamic Mechanism and Data Augmentation

Second author | 44th Chinese Control Conference (CCC 2025)

- Co-developed an information-entropy data augmentation method (KDE with L_1 sparsity) and an Elastic Net error-compensation model; benchmarked against MLP/SVR/LSTM on CSTR simulation (MAE/RMSE/ R^2).

Technical Projects

AgentKey – Portable AI Coding-Agent Runtime

Open-source | 2026

Stack: Shell / MCP / OpenAI-compatible / DeepSeek

- A USB-ready runtime that carries a personal Codex-style AI coding agent (encrypted provider setup, skills, memories, and MCP config) across any Mac or Linux machine.

HazLab Copilot – Evidence-Grounded Lab Safety Agent

Open-source | 2026

Stack: Next.js / TypeScript / Qwen / Gemma LLM

- An LLM agent for laboratory safety (chemical management, inspection, emergency response) with retrieval-based evidence grounding and a HazLab-Bench evaluation suite.

AI Closet – Vision-LLM Wardrobe Assistant

Open-source | 2026

Stack: Next.js / React / TypeScript / Vision LLM

- A Vision-LLM pipeline that parses garment images into structured tags (category/color/style/occasion) and recommends outfits with preference-feedback re-ranking.

Internship

Bluestar Intelligent Cloud (Shandong) Technology Co., Ltd. – Beijing Branch Jun 2025 – Aug 2025

Technical Intern, Production Digitalization Dept. (Industrial Software Development)

- Contributed to industrial software development for smart-manufacturing scenarios: requirement analysis, feature development, integration testing, and documentation.

Technical Skills

Programming Python (proficient), C/C++ (embedded), SQL.

LLMs Transformer/GPT/LLaMA architectures; SFT/RLHF/DPO fine-tuning; Vision-LLM integration and ASR → LLM → TTS pipelines.

ML / Data Scikit-learn (large-scale classification/calibration/evaluation), PyTorch (training & debugging).

Frameworks PyTorch, HuggingFace Transformers, NumPy/Pandas/Matplotlib, Git, Linux, Arduino/ESP-IDF.

Languages English (fluent, English-taught program), Mandarin (native).